do…while loop in Java

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A do…while loop in Java is a control structure that runs a block of code at least once, and then repeats it as long as a given condition remains true. It is called an "exit-controlled" loop because the condition is checked at the end of the loop.

Syntax:

```
do{
   //loop body: code to execute
}while(condition);
```

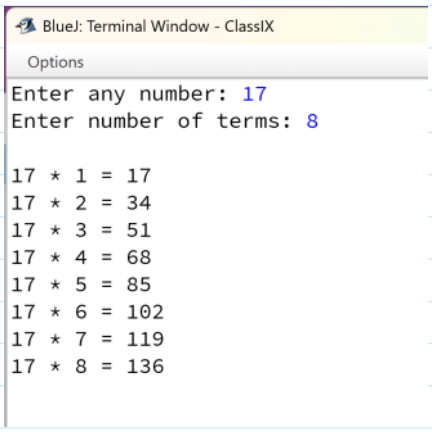
← body runs first.
← Then condition is evaluated.
← If condition is true, the loop repeats; if false, the loop exits.
The semicolon after while(condition) is mandatory.

Execution order:
1. Execute code in do{...}.
2. Evaluate condition.
3. If true, go back to step 1; if false stop.

```
package Looping.dowhileLoop;
import java.util.*;


/**
 * Write a description of class DoWhileExample1 here.
 *
 * @author (your name)
 * @version (a version number or a date)
 */
public class DoWhileExample1
{
    public static void main(String[] args){
        //multiplication table of a number entered by the user
        Scanner sc = new Scanner(System.in);
        System.out.print("\f");
        int num, start = 1, end;
        System.out.print("Enter any number: ");
        num = sc.nextInt();
        System.out.print("Enter number of terms: ");
        end = sc.nextInt();
        do{
            System.out.print("\n"+num + " * " + start + " = " + (num*start));
            start++;
        }while(start <=end);
                    T
    }
}
```

Output:



The do…while(); loop is mainly used for data validation.

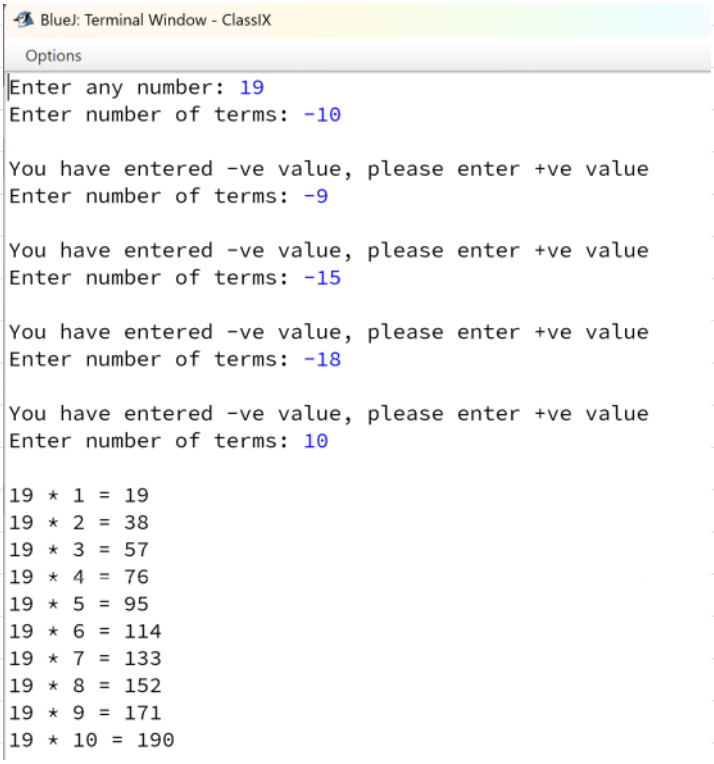
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        //multiplication table of a number entered by the user
        Scanner sc = new Scanner(System.in);
        System.out.print("\f");
        int num, start = 1, end;
        do{
            System.out.print("Enter any number: ");
            num = sc.nextInt();
            if(num <= 0){
                System.out.println("\nYou have entered -ve value, please enter +ve value");
            }
        }while(num <= 0);

        do{
            System.out.print("Enter number of terms: ");
            end = sc.nextInt();
            if(end <= 0){
                System.out.println("\nYou have entered -ve value, please enter +ve value");
            }
        }while(end <= 0);

        do{
            System.out.print("\n"+num + " * " + start + " = " + (num*start));
            start++;
        }while(start <=end);
    }
}
```

Output:



WAP to input a positive int value and find its sum of digits using do…while loop.
Example:
Input: Enter any number: 273
Output: Sum of digits: 12

Homework:
Convert the for loop programs in do…while loop.